

Project Summary

Overview

With demand for wireless network capacity growing year over year, the shift to millimeter-wave (mmWave) frequencies and beyond has amplified real-world challenges in the planning and deployment of indoor systems such as distributed antenna systems (DAS). Traditional approaches retrofit existing structures through strategic antenna and radio placement. While serviceable at sub-6 GHz frequencies, where propagation and reflection are more tolerant of obstacles and material variation, these methods become problematic at mmWave due to channel sparsity, blockage sensitivity, diffuse reflections, and the need for precise beam alignment.

Beamforming introduces a shift from broad-area transmission to narrow, highly directional paths. This increases link efficiency but removes many of the propagation redundancies that made earlier systems resilient to obstructions and layout variability. At mmWave frequencies, reliable communication depends on the environment's ability to preserve signal energy through line-of-sight propagation or strong specular reflections. These conditions motivate a reevaluation of how interior spaces are structured and how materials are selected.

This project proposes a design approach based on modular, pre-characterized layout elements tailored for high-frequency wireless performance. Components such as corridors, corners, and transitions are treated as tileable design units and paired with materials that are strategically positioned and shaped to support reflection or transmission at key locations. This framework supports the construction of architecturally functional floorplans that are inherently compatible with directional wireless coverage, reducing the need for costly retrofits and improving performance in dense indoor environments.

Intellectual Merit

This project develops a design methodology for mmWave propagation in reflective indoor environments by analyzing how geometry, material selection, and spatial placement affect signal redirection and coverage. It combines electromagnetic theory, ray tracing, and system-level modeling to evaluate modular interior elements tailored for directional beam support. These insights enable the creation of floorplans that preserve beam continuity and reduce non-line-of-sight loss. The results inform the design of interior spaces and material usage for environments operating at mmWave frequencies, and the methodology may be extended to sub-terahertz bands where passive reflection remains critical due to limited diffraction, low penetration, and highly directional communication.

Broader Impacts Of The Proposed Work

By enabling cost-effective and energy-efficient indoor mmWave coverage, this project has the potential to improve wireless capacity in high-demand environments such as offices, factories, and public venues. The proposed design framework may reduce infrastructure costs and environmental impact by optimizing layout and material usage to minimize the number of required access points. These improvements support scalable, high-performance wireless deployments and promote sustainable practices in future indoor network design.

Enhancing Indoor MmWave Coverage Through Passive Reflector Design And Placement Optimization Using Ray-Tracing Simulations

Introduction

The rapid evolution of wireless communications, driven by bandwidth-intensive applications such as immersive augmented reality, tactile internet, and smart industrial systems, has ushered in an urgent need for high-capacity indoor wireless infrastructure. To meet this demand, research and regulatory communities have turned to millimeter wave (mmWave) and sub-terahertz (THz) frequencies (30–300 GHz and 100 GHz–3 THz, respectively), where vast underutilized spectrum offers the potential for multi-gigabit data rates. In 2019, the U.S. Federal Communications Commission (FCC) opened up spectrum above 95 GHz for unlicensed use [1], reinforcing momentum for developing next-generation wireless technologies. Simultaneously, the IEEE 802.15.3d standard has been established to enable global wireless communication in the 250–350 GHz range, targeting data rates up to 100 Gbps [2].

However, these frequency bands are not without challenges—particularly indoors, where propagation suffers from significant free-space path loss, limited penetration through walls, and sensitivity to small-scale surface characteristics [3]. Recent empirical studies have highlighted the complex behavior of mmWave signals in indoor environments, demonstrating that the dominant propagation mechanisms at these frequencies—reflection, scattering, and transmission—are profoundly influenced by material permittivity, surface roughness, and angle of incidence [4, 5].

For example, measurements at 28, 73, and 142 GHz show that the reflection loss from drywall decreases with increasing frequency and grazing angle, suggesting stronger specular reflection at higher frequencies [3]. This is complemented by findings that partition loss increases with frequency and is highly dependent on polarization configuration and material type, with clear glass and drywall exhibiting significant depolarization effects at 140 GHz [3]. Simultaneously, scattering cannot be ignored at mmWave and THz frequencies due to the comparable scale of wavelength and surface roughness. Anioke et al. [4] demonstrated, through a modified Saleh-Valenzuela (SV-UWB) model, that rough materials like brick and concrete scatter energy away from the specular direction, reducing received power and modifying the channel impulse response.

To overcome these challenges, this project proposes going further than the retrofit of existing indoor spaces by creating tools to assist in the design and layout of interior space which are better suited to the needs of wireless systems operating at mmWave frequencies, with an eye toward eventual use at sub-THz and THz bands. The approach focuses on building a set of simple, repeatable layout modules such as corridor segments, junctions, and transition zones. These modules are selected based on known propagation behaviors like reflection and diffusion. They are intended for use during the planning phase of building construction or renovation, allowing coverage and beam preservation to be considered before walls are placed and materials are chosen.

The core research effort will center on designing and validating a set of standardized interior layout modules with predictable mmWave propagation behavior. Using Wireless InSite and other professional tools, we will simulate and characterize signal paths and behavior across different layouts and materials to evaluate beam preservation, NLOS support, and coverage consistency. These results will inform a modular design library, where each element—such as corridor elbows or wall transitions—comes with documented performance characteristics. The final deliverable will be a lightweight design tool that allows engineers or architects to assemble floorplans using these

modules and receive real-time propagation estimates. This system is intended to support early-stage planning for mmWave deployments in offices, venues, and other complex indoor environments.

Background And Related Works

Millimeter wave (mmWave) communication has emerged as a key enabler for future indoor wireless systems due to its potential for extremely high data rates enabled by wide bandwidth availability above 24 GHz. However, the propagation characteristics of mmWave signals—particularly in indoor and non-line-of-sight (NLOS) scenarios—present several challenges. These include high path loss, susceptibility to blockages, and limited diffraction, which can severely degrade connectivity. Recent research has explored the properties of mmWave indoor propagation as well as various methods to enhance coverage, including the use of passive and intelligent reflecting surfaces. This section reviews the state-of-the-art in mmWave indoor propagation modeling and coverage enhancement using reflectors.

Extensive measurement campaigns at 28, 73, and 142 GHz have revealed that indoor mmWave and sub-THz frequencies exhibit distinct propagation behaviors compared to lower microwave bands [6]. Reflection dominates over scattering in environments with smooth surfaces like dry-wall, with reflection losses decreasing as frequency increases. For instance, drywall reflection loss at 80° incident angle decreases from -3.18 dB at 28 GHz to -0.36 dB at 142 GHz, while permittivity increases from 4.7 to 6.4 across the same frequencies. Additionally, drywall and clear glass exhibit frequency-dependent partition loss and depolarization effects, with substantial absorption losses at 142 GHz.

Moreover, the study confirms that classical Close-In (CI) path loss models remain valid up to 142 GHz. Path loss exponents (PLEs) for line-of-sight (LOS) were found to be modestly increasing with frequency: 1.7 at 28 GHz, 1.6 at 73 GHz, and 2.0 at 142 GHz. Importantly, NLOS PLEs remained similar or slightly better at higher frequencies due to increased reflection strength, suggesting opportunities for improved indoor NLOS coverage using reflective surfaces.

While specular reflection dominates in smooth surfaces, indoor materials with rough textures can induce significant scattering at mmWave frequencies. A modified Saleh-Valenzuela (SV) ultrawideband model incorporating scattering effects reveals that surface roughness, grazing angle, and frequency are critical in determining scattering behavior [7]. For rough materials like brick and concrete, scattering can substantially affect the mean received signal energy—ranging up to 48 dB depending on material and incidence angle. The scattering coefficient, derived from Rayleigh roughness and reflection loss models, is found to be non-negligible, especially around critical grazing angles (e.g., 23° to 40° for typical materials). These findings reinforce the need for accurate modeling of both reflection and scattering in mmWave indoor environments, particularly for robust channel estimation and reflector placement.

Highly effective mmWave reflection in NLOS environments relies on materials with strong specular reflectivity and effective beam energy preservation. Experimental results by Khawaja et al. demonstrated that flat aluminum sheets, oriented at appropriate azimuth angles, can reflect 28 GHz mmWave signals with received power levels approaching line-of-sight (LOS) conditions when the reflector surface area matches or exceeds the beam footprint [8]. These findings support the use of polished, conductive materials, such as aluminum or coated metal panels, for redirecting narrow mmWave beams efficiently. To explore alternative materials that are easier to fabricate or integrate into indoor infrastructure, P’erez-Escribano et al. characterized a range of polymers commonly used in 3D printing [9]. Among these, high-impact polystyrene (HIPS) showed the most favorable properties, with low loss tangent and stable dielectric behavior up to 67 GHz. These results suggest that HIPS and similar polymers may offer a potentially lower-cost alternative to

metal for constructing lightweight or modular reflectors, especially in environments where ease of fabrication or rapid prototyping is prioritized.

In contrast to specular reflectors, some materials naturally scatter rather than reflect incident mmWave energy in a single direction. A controlled degree of diffuse scattering can be useful in areas where precise beam alignment is impractical but moderate angular spread is needed to extend coverage into shadowed or offset regions. Ferreira et al. reviewed a range of common indoor construction materials—including brick, concrete, chipwood, and plasterboard—and found that their rough surfaces and dielectric properties contribute to non-specular behavior at mmWave frequencies [10]. These materials typically exhibit relative permittivity values between 4 and 6 [10], a range that enables measurable reflection at mmWave frequencies while still promoting broad angular dispersion due to surface roughness. Their irregular textures, often comparable in scale to mmWave wavelengths, cause energy to scatter over a wider angular range. This situation highlights a broader challenge that motivates our work. After beam energy is preserved through controlled reflection, it must often be redirected into nearby rooms or occluded spaces where coverage is weak. The ideal amount of scattering depends on the geometry and layout of the space. If too much energy is diffused, it may be wasted, while too little may leave critical regions under-served.

Reflector performance depends not only on material properties but also on geometric configuration. Khawaja et al. analyzed the relationship between reflector size, orientation, and received power using indoor measurements supported by analytical modeling [8]. Their results show that once the reflector area exceeds the core mmWave beam footprint, further increases in surface area produce minimal additional gain. For example, a $0.61\text{ m} \times 0.61\text{ m}$ flat aluminum sheet was sufficient to achieve signal strength near free-space levels at the receiver, while larger reflectors provided only marginal improvements. Proper orientation was also essential. Maximum signal redirection occurred when the transmitter and receiver boresights were symmetrically aligned with the reflector’s surface normal. These results establish practical limits on reflector scale and placement for indoor environments, emphasizing that careful alignment is often more important than raw surface area when redirecting directional mmWave energy.

Geometries designed for spatial coverage also play a critical role in shaping how mmWave energy is distributed within indoor environments. Curved reflectors scatter incident energy over wider angles due to their geometry, which redirects the beam away from the specular axis. Khawaja et al. evaluated both spherical and cylindrical reflectors and found that these geometries help distribute energy across broader receiver regions by inducing beam divergence in azimuth and elevation planes [8]. Cylindrical reflectors, in particular, produced more consistent power levels due to their more uniform angular spread along one axis. Their analytical models of effective area and reflection behavior confirm that such curved geometries can improve spatial coverage in scenarios where robustness is prioritized over directional precision. This tradeoff complements Ferreira et al.’s characterization of rough indoor surfaces, which scatter energy broadly rather than focusing it [10]. Together, these results suggest that carefully shaped geometries, when used to distribute preserved beam energy more broadly, can improve both continuity along intended paths and coverage in spatially offset regions within complex mmWave layouts.

To mitigate high path loss and limited diffraction in mmWave indoor NLOS scenarios, passive metallic reflectors have been proposed as a low-cost, energy-efficient solution. Experiments at 28 GHz demonstrate that flat square reflectors can yield up to 20 dB signal gain in indoor corridors, with the ability to steer reflected energy by adjusting azimuth and elevation angles [11]. However, gains beyond a certain reflector size plateau, suggesting diminishing returns with increasing size.

Ray tracing simulations validate these experimental findings, showing that passive reflectors effectively redirect energy into shadowed regions with minimal complexity. The optimal orientation and size depend on the incident beam directionality and placement within the indoor environment.

Advanced implementations such as MilliMirror introduce configurable, passive reflectors composed of metal-backed dielectric cuboids that introduce phase shifts to incident mmWave signals [12]. These surfaces enable beam shaping and steering, effectively acting as static, programmable reflectors. By optimizing dielectric thickness across sub-wavelength unit cells, MilliMirror achieves wide angular coverage and sustained signal strength in complex NLOS scenarios (e.g., around-the-corner links). While offering greater control and performance than flat metallic sheets, the challenge of beam squinting due to frequency dependence remains, prompting the need for wideband design strategies.

Beyond fully indoor use, transmissive reconfigurable intelligent surfaces (RISs) have been proposed for enhancing outdoor-to-indoor (O2I) mmWave coverage, where wall penetration loss is a critical bottleneck. A 3D system-level analysis showed that deploying multiple small-scale RISs at carefully optimized locations on the building envelope can improve indoor coverage by up to 69.5% compared to a single large RIS [13]. Coverage probability formulations accounting for blockage density, spatial correlation, and Nakagami fading provide insights for optimal RIS placement in complex environments.

Finally, ray tracing studies confirm that replacing drywall with metallic wall coatings significantly improves mmWave signal propagation indoors [14]. At 60 GHz, connectivity improves from 86% to 97.5% with reflective surfaces. The improvement is more pronounced at higher frequencies due to increased path loss and the higher reflectivity of metallic surfaces. These findings suggest that building materials and surface coatings can play a pivotal role in the feasibility of indoor mmWave deployments.

Collectively, the above findings support the hypothesis that integrating strategically placed reflectors—both passive and intelligent—can substantially enhance mmWave indoor coverage, particularly in NLOS conditions. Smooth high-permittivity materials and reflectors enable stronger reflection paths, while rougher surfaces introduce useful scattering. Ray tracing tools combined with empirical modeling guide optimal reflector design and placement. These techniques are especially relevant for 6G and beyond systems operating above 100 GHz, where signal robustness and energy efficiency become critical.

Proposed Study

Objectives And Significance

This project aims to develop a modular design framework for indoor environments that supports predictable mmWave connectivity using spatial layouts constructed from pre-characterized geometric building blocks. Instead of adding reflectors or diffusers after construction, we treat signal propagation as a primary design constraint during the architectural planning process. The central objective is to create a library of corridor segments, junctions, and transitional elements that exhibit known propagation behaviors, such as specular reflection, controlled scattering, or angular diffusion, at mmWave frequencies. These modules will be used to assemble complete interior layouts that maintain strong signal paths, redirect energy around obstacles, and extend coverage into occluded or non-line-of-sight regions. The final outcome is a lightweight design tool or planning framework that enables users to create indoor layouts with favorable mmWave coverage, without requiring deep expertise in wireless system design.

Relation To Long-Term Goals And Current Knowledge

This work supports the long-term vision of rethinking indoor wireless infrastructure to support highly directional communication at mmWave, sub-THz, and THz frequencies. As bandwidth

demands grow and wireless systems evolve toward higher frequency bands, conventional in-building strategies like dense access point placement or legacy DAS retrofits become increasingly impractical. However, the opportunity to engineer interior spaces to support directional wireless propagation has received limited attention in the context of early-stage spatial planning. This project proposes a modular approach to building design where corridor and wall elements are selected for their known propagation behavior and arranged to preserve, guide, or diffuse directional beams. By combining empirical ray tracing with data-driven analysis of layout performance, we aim to deliver a practical framework for early-stage floorplan planning that supports efficient, low-power wireless coverage at emerging frequency bands.

While the immediate focus is on modular elements that can be explicitly modeled and placed, such as ceiling-mounted reflectors or diffusers, this framework is intended to lay the groundwork for more integrated design strategies. In future work, we hope to explore how architectural features—angled walls, ceiling contours, or geometric motifs—can be shaped to guide wireless energy while contributing to the visual identity of a space. This direction seeks to blend form and function, enabling high-frequency propagation behavior to become an intentional and expressive part of interior architecture rather than a concealed afterthought.

Relation To Ongoing Work By PI And Others

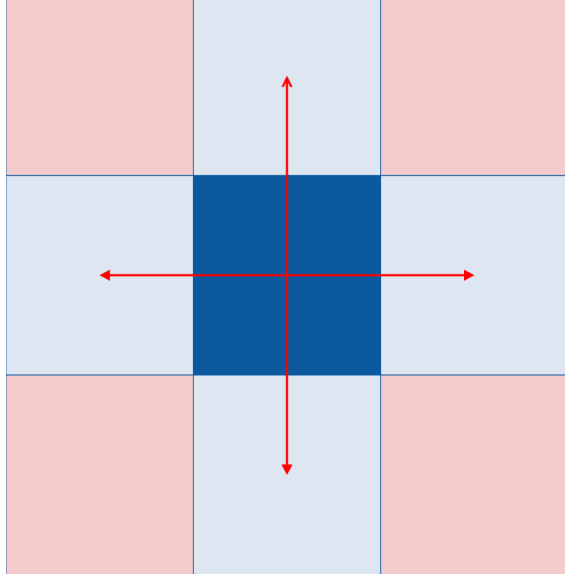
This project builds on related experience in mmWave systems engineering, including material characterization, reflector performance evaluation, and real-world testbed development. These efforts provide a strong foundation for understanding the challenges of directional communication in indoor environments and the limitations of reactive deployment strategies. In contrast to prior work that explores passive reflectors and reconfigurable intelligent surfaces (RIS) as post-construction enhancements, this project investigates how indoor wireless performance can be shaped during the architectural design process itself. It draws on adjacent wireless expertise to initiate a new line of inquiry at the intersection of communication engineering and spatial design.

General Plan Of Work And Methodology

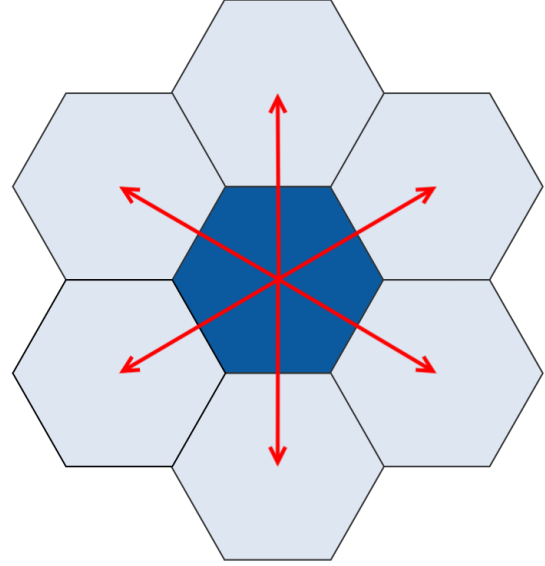
The proposed research is divided into four distinct phases, each addressing a specific technical question. The investigation begins with the evaluation of tiling strategies for modular indoor layouts, followed by the characterization of passive components such as reflectors and diffusers. In this context, a diffuser refers to a surface or geometric element that intentionally scatters mmWave energy over a wider angular range to reach occluded or spatially offset regions. These elements are then analyzed through ray tracing to understand their impact on mmWave propagation. The final phase focuses on developing a lightweight planning tool that supports layout exploration with minimal RF expertise.

Phase 0: Evaluating Tile Strategies for Modular Indoor Design

This foundational phase defines the geometric basis for modular indoor design by evaluating how different tile shapes affect beam directionality and layout flexibility. This preliminary investigation examines how tiles such as squares and hexagons influence both propagation support and design versatility. As illustrated in Figure 1a, square-based layouts restrict mmWave beam paths to orthogonal angles, simplifying layout assembly but limiting angular coverage. In contrast, Figure 1b shows that hexagonal tiling supports a richer set of beam orientations. However, when applied to rectangular or curved boundaries such as office floors or event centers, hexagonal layouts introduce spatial inefficiencies and edge handling complications, as demonstrated in Figure 2. The goal of this phase is to identify a tiling strategy that best balances support for directional beam paths,

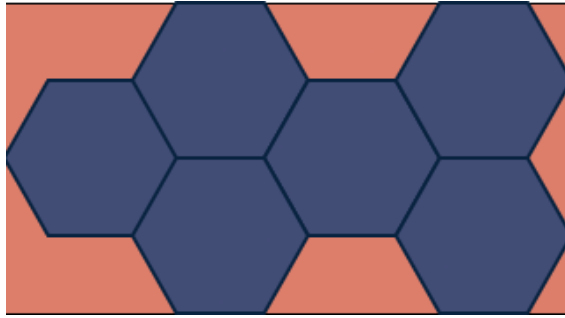


(a) Beam directions in square-tiling layout.

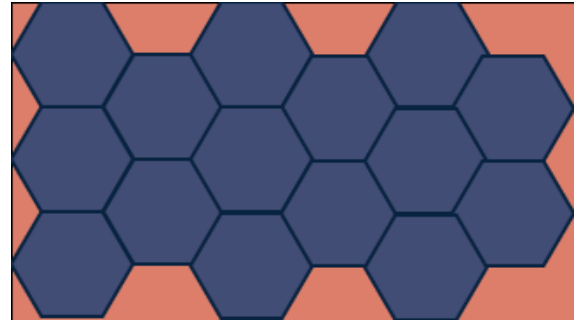


(b) Beam directions in hexagonal-tiling layout.

Figure 1: Comparison of directional constraints imposed by square and hexagonal tile layouts.



(a) Larger hexagon tiling a rectangular space.



(b) Smaller hexagon tiling a rectangular space.

Figure 2: Tiling inefficiencies and edge handling in hexagonal grids applied to rectangular spaces.

coverage across varied floorplan geometries, ease of modular assembly, and freedom for architectural expression.

Phase 1: Propagation Characterization with Reflectors and Diffusers The first major phase of the project focuses on ray tracing and propagation characterization in representative indoor environments. Using Wireless InSite or equivalent electromagnetic simulation tools, we will model canonical building modules—such as corridors, junctions, elbows, and room transitions—constructed from selected tile geometries. Each configuration will be evaluated in terms of beam preservation, reflection strength, angular spread, and non-line-of-sight (NLOS) redirection behavior.

An example scenario is shown in Figure 4, where a transmitter located in a corridor uses a strategically placed reflector to guide mmWave energy into an adjacent room. A diffuser at the room entrance scatters the redirected energy across a wide angular range, enhancing uniform spatial coverage. This scene is simulated by running Remcom’s publicly available mmWave indoor modeling project in Wireless InSite, without modification, to illustrate representative ray interactions

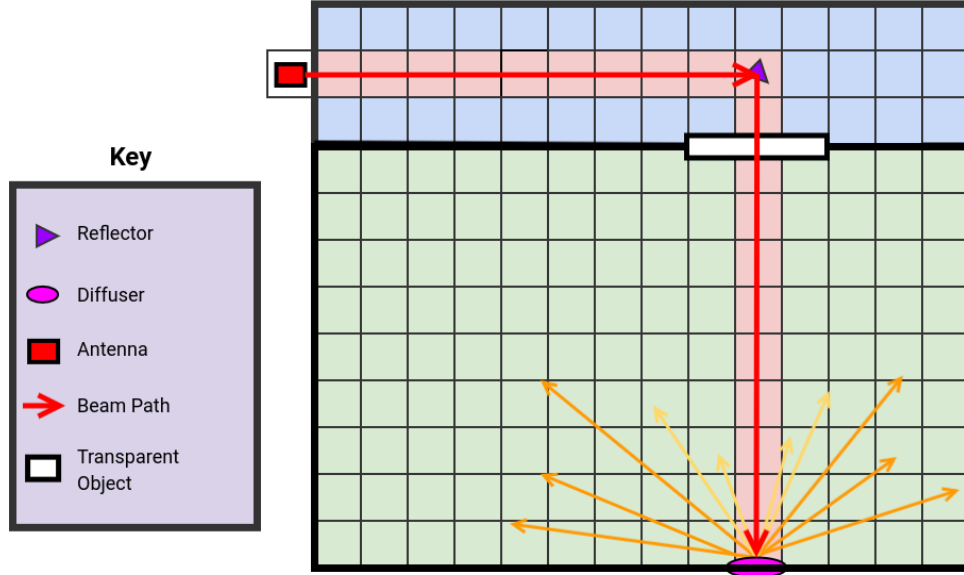


Figure 3: Conceptual layout demonstrating mmWave signal propagation using a ceiling-mounted reflector and diffuser. The beam is redirected into a reflector and diffused into a room with occluded line-of-sight.

and propagation behaviors [15]. The visualization illustrates the proposed approach of combining reflective steering and diffusive scattering to overcome the challenges of high-frequency NLOS coverage in dense indoor layouts.

The simulation results will contribute to a modular design library that captures the propagation characteristics of different reflector-diffuser configurations. This library will underpin a lightweight planning toolkit for mmWave-aware architectural design. The toolkit will allow users—such as building designers or RF system integrators—to compose complex indoor layouts with known electromagnetic behavior, without requiring deep wireless expertise.

Phase 2: Ray Tracing and Environment Modeling

Phase 2 focuses on modeling complete indoor environments constructed within the tiling scheme selected in Phase 0. Using Wireless InSite or equivalent tools, we will simulate composite layouts such as corridors, junctions, room transitions, and open areas, incorporating reflector and diffuser elements as defined in Phase 1. Simulation outputs will be used to evaluate layout-scale performance metrics such as path continuity, beam dispersion, and NLOS coverage. An example configuration is shown in Figure 3, where a beam is redirected by a ceiling-mounted reflector and diffused into an occluded room. To assist in refining element placement and configuration, we will employ machine learning models trained on ray tracing data to identify spatial arrangements that improve directional stability and overall signal distribution. These models will serve as a recommendation layer to guide the selection and placement of passive elements in complex environments under practical architectural constraints.

Phase 3: Planning Toolkit for Propagation-Aware Layout Design

Phase 3 focuses on the development of a lightweight planning toolkit that allows users to assemble indoor layouts using the pre-characterized modules defined in earlier phases. The toolkit will

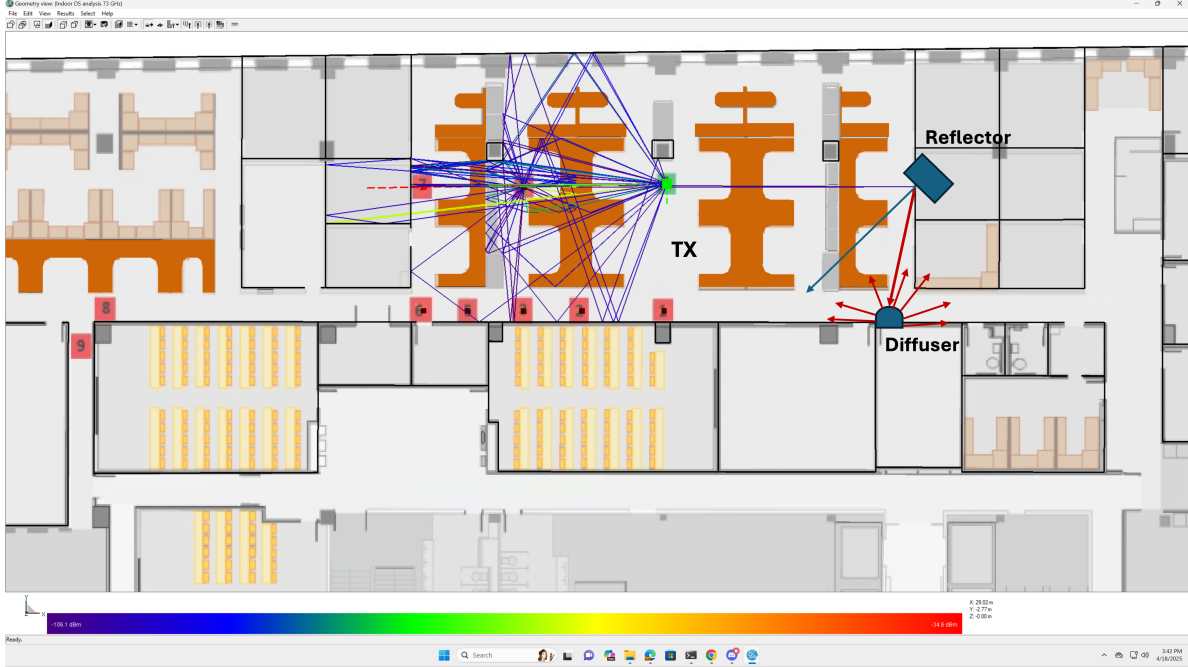


Figure 4: Ray tracing simulation showing beam steering via a reflector and spatial dispersion via a diffuser to enhance indoor mmWave coverage.

provide a library of geometric and functional components, including corridor segments, junctions, and transition modules, each annotated with their corresponding propagation behaviors. Users will be able to explore different floorplan configurations and receive automated feedback on expected coverage, beam path continuity, and NLOS support. The system will be designed for use during the early stages of architectural planning and will not require specialized RF expertise. This phase will also explore interaction models that support intuitive layout assembly, with an emphasis on design patterns that are broadly familiar to non-expert users, such as component snapping, visual alignment cues, and direct manipulation interfaces. Rather than simulating raw electromagnetic behavior, the toolkit will operate on abstracted results from prior ray tracing, allowing fast, layout-driven evaluations that guide design decisions for mmWave-capable interior spaces.

Evaluation And Success Metrics

The success of the proposed research will be evaluated based on how effectively the resulting system enables users to design indoor layouts that support directional mmWave communication. This includes both beam preservation through the use of reflectors and area coverage through the use of diffusers, as well as how well these modeled behaviors correspond to real-world signal propagation outcomes. Specific evaluation criteria will include:

- **Design Flexibility:** The ability of the selected tiling strategy and module set to support a wide range of floorplan configurations and interior design goals without excessive constraint.
- **Beam Engineering Flexibility:** The degree to which reflectors and diffusers can be selectively applied to shape directional paths and NLOS coverage in varying spatial contexts.
- **Beam Preservation Performance:** Demonstrated ability to maintain directional energy through multi-module corridor configurations with minimal angular loss.

- **Area Coverage Performance:** Improved angular dispersion and signal uniformity into occluded or partitioned spaces when diffusers are appropriately placed.
- **Infrastructure Efficiency:** Number of access points required to meet coverage and connectivity targets, measured in environments with and without passive components. This metric will evaluate how the use of reflectors and diffusers affects overall access point requirements.
- **Predictive Accuracy of Abstracted Models:** Agreement between ray-tracing-informed performance estimates and reference measurements in analogous physical layouts, used to validate the effectiveness of the simulation-to-design abstraction.

Broader Impacts Of The Proposed Work

Reliable mmWave coverage indoors remains a challenge. This project offers a layout-centered solution that reduces cost, lowers energy use, and expands the design space for future wireless deployments. Reflectors and diffusers embedded into ceilings, corners, or structural elements can reduce the number of access points needed to maintain coverage and beam continuity. This enables lower infrastructure costs and long-term energy savings while preserving architectural flexibility.

Unlike outdoor systems, indoor environments offer intentional control over both geometry and material selection. This project treats the built environment not as an obstacle to overcome, but as an asset to be configured. By shaping walls, corridors, and surface interactions to favor signal continuity and redirection, the system addresses challenges such as sparse connectivity, signal blockage, and poor angular coverage in occluded spaces. This approach positions wireless behavior as something that can be shaped through spatial design, enabling performance outcomes to be guided through controllable layout decisions.

The toolkit developed in this project promotes adaptability and long-term scalability by enabling layout evaluation based on expected mmWave performance, without requiring specialized RF expertise. By abstracting ray-tracing results into usable layout metrics, the system can support both new construction and retrofitting efforts. In doing so, this work encourages planning approaches that integrate wireless performance into the design process.

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